I. System jumper & connectors

X1 : 2xRS232

1 – RX input RS232 -> MAX -> RX STM32 PA10 - 43
2 – TX output RS232 <- MAX <- TX STM32 PA9 - 42
3 – GND
4 – RX input RS232 -> MAX -> & TX STM32 PC6 - 37 -> RX EP1C6 - 119; X13 - 2
5 – TX output RS232 <- MAX <- RX STM32 PC7 - 38 <- TX EP1C6 - 113; X13 - 3

X2 : RS485

1 – RS485 - A
2 – RS485 - B
3 – GND
4,5 – RS485 120 Om enable jumper

X3 : CAN

1 – CAN - L
2 – CAN - H

X4 : CAN 120 Om enable jumper

X5 : Microcontroller BOOT mode (1-2:PIN PC13 BOOT) or (2-3:BOOT0=1)

1 – PC13 pin
2 – BOOT0 pin
3 – VDD 3.3V

X6 : Power external input +5V

1 - GND
2 - +5V Board

X7 : USB Type-C Power Jumper

1 - +5V Type-C
2 - +5V Board

X8 : +5V output from USB Type-C

1 - +5V Type-C
2 - GND

X9 : Watchdog timer jumper (1-2:WDTON) or (2-3:WDTOFF)

1 - Reset WDT & Button
2 - Reset
3 - Reset Button

X10 : Processor BOOT mode enable jumper

X11 : Speaker Output

1 - OUT_M
2 - OUT_P

X12 : V3S VRTC (1-2:VRTC = VDD 3.3V) or (2-3:VRTC=GND)
1 – VDD 3.3V
2 – V3S VCC-RTC - 98
3 – GND

X13 : IN-OUT
1 – V3S - 100
2 – EP1C6 - 119
3 – EP1C6 - 113

X14 : BAT Power
1 - +V BAT
2 - GND

II. Pin Connectors

Y1 : STM32F412
01 - STM32F412[61] - PB8 - I2C1_SCL / TIMER4_CH3 / CAN1_RX
02 - STM32F412[40] - PC9 - MCO2
03 - STM32F412[62] - PB9 - I2C1_SDA / TIMER4_CH4 / CAN1_TX
04 - STM32F412[24] - PC4 - ADC1_IN14
05 - STM32F412[28] - PB2 - BOOT1
06 - STM32F412[22] - PA6 - ADC1_IN6 / TIMER3_CH1
07 - STM32F412[27] - PB1 - TIMER3_CH4
08 - STM32F412[21] - PA5 - ADC1_IN5 / TIMER2_CH1
09 - STM32F412[26] - PB0 - ADC1_IN8 / TIMER1_CH2N / TIMER3_CH3
10 - STM32F412[20] - PA4 - ADC1_IN4 / UART2_CK
11 - STM32F412[36] - PB15 - SPI2_MOSI / TIMER1_CH3N
12 - STM32F412[10] - PC2 - ADC1_IN12 / SPI2_MISO
13 - STM32F412[29] - PB10 - SPI2_SCK / TIMER2_CH3
14 - STM32F412[09] - PC1 - ADC1_IN11
15 - VDD_3.3V
16 - STM32F412[08] - PC0 - ADC1_IN10
17 - GND
18 - STM32F412[15] - PA1 - ADC1_IN1 / TIMER2_CH2 / TIMER5_CH2
19 - STM32F412[35] - PB14 - SPI2_MISO / TIMER1_CH2N
20 - STM32F412[14] - PA0 - ADC1_IN0 / TIMER2_CH1 / TIMER5_CH1

Y2 : SYSTEM
01 - V3S[49] - UART0_RX - Y4[44]
02 - GD32F103[21] - PA9 - USART0_TX
03 - V3S[48] - UART0_TX - Y4[42]
04 - GD32F103[22] - PA10 - USART0_RX
05 - VDD_3.3V
06 - GD32F103[25] - PA13 - SWDIO
07 - NC
08 - GD32F103[28] - PA14 - SWCLK
09 - RESET
10 - GND

Y3 : CMSYS-DAP
01 - VDDIN
02 - GD32F103[13]-PA6 - nRST   control
03 - GD32F103[12]-PA5 - TDO   control
04 - GD32F103[11]-PA4 - TDI   control
05 - GD32F103[08]-PA1 - SWDIO control
06 – STM32F412[46]-PA13- SWDIO in
07 - GD32F103[07]-PA0 - SWCLK control
08 – STM32F412[49]-PA14- SWCLK in
09 - VDD_3.3V
10 - GND

Y4 : COMMON
01 - GND
02 - VDD_5.0V
03 - V3S[15] - LCD_D18
04 - V3S[14] - LCD_D19
05 - V3S[13] - LCD_D20
06 - V3S[11] - LCD_D21
07 - V3S[07] - LCD_D22
08 - V3S[06] - LCD_D23
09 - V3S[24] - LCD_D10
10 - V3S[23] - LCD_D11
11 - V3S[22] - LCD_D12
12 - V3S[18] - LCD_D13
13 - V3S[17] - LCD_D14
14 - V3S[16] - LCD_D15
15 - V3S[33] - LCD_D2
16 - V3S[32] - LCD_D3
17 - V3S[31] - LCD_D4
18 - V3S[30] - LCD_D5
19 - V3S[28] - LCD_D6
20 - V3S[27] - LCD_D7
21 - V3S[37] - LCD_CLK
22 - V3S[35] - LCD_HS
23 - V3S[34] - LCD_VS
24 - V3S[36] - LCD_DE
25 - EP1C6[129] - IO.0
26 - V3S[39]   - PB0-UART2_TX
27 - EP1C6[130] - IO.1
28 - V3S[40]   - PB1-UART2_RX
29 - EP1C6[131] - IO.2
30 - V3S[41]   - PB2-UART2_RTS
31 - EP1C6[132] - IO.3
32 - V3S[42]   - PB3-UART2_CTS
33 - EP1C6[142] - IO.4
34 - V3S[43]   - PB4-PWM0
35 - EP1C6[143] - IO.5
36 - V3S[43]   - PB5-PWM1
37 - EP1C6[144] - IO.6
38 - V3S[45]   - PB6-SCL0
39 - EP1C6[01]   - IO.7
40 - V3S[46]   - PB6-SDA0
41 - EP1C6[02]   - IO.8
42 - V3S[48]   - PB8-UART0_TX

43 - EP1C6[03] - IO.9
44 - V3S[49] - PB9-UART0_RX
45 - EP1C6[04] - IO.10
46 - V3S[09] - PE21-SCL1
47 - EP1C6[05] - IO.11
48 - V3S[08] - PE22-SDA1
49 - ADS1248[06] - REFN0
50 - ADS1248[05] - REFP0
51 - ADS1248-R - ERTDG
52 - ADS1248-R - ERTD
53 - ADS1248[11] - IN0
54 - VDDIN
55 - ADS1248[13] - IN4
56 - ADS1248[12] - IN1
57 - ADS1248[15] - IN6
58 - ADS1248[14] - IN5
59 - ADS1248[20] - IEXC1
60 - ADS1248[16] - IN7
61 - ADS1248[17] - IN2
62 - ADS1248[19] - IEXC2
63 - ADS1248[18] - IN3
64 - AGND

Y5 : JTAG
01 - JTAG_TCK - EP1C6[88]
02 - GND
03 - JTAG_TDO - EP1C6[90]
04 - nCONFIG - EP1C6[14]
05 - JTAG_TMS - EP1C6[89]
06 - CONFDONE - EP1C6[86]
07 - STATUS - EP1C6[87] - GD32F103[16]
08 - DATA - EP1C6[13] - GD32F103[34]
09 - JTAG_TDI - EP1C6[95]
10 – DCLK - EP1C6[24] - GD32F103[33]

Y6 : HUB-75
01 - EP1C6[99] - IO.34
02 - EP1C6[100] - IO.35
03 - EP1C6[96] - IO.33
04 - GND
05 - EP1C6[97] - IO.32
06 - EP1C6[85] - IO.31
07 - EP1C6[84] - IO.30
08 - EP1C6[83] - IO.29
09 - EP1C6[82] - IO.28
10 - EP1C6[79] - IO.27
11 - EP1C6[78] - IO.26
12 - EP1C6[77] - IO.25 - LED WHITE
13 - EP1C6[76] - IO.24
14 - EP1C6[75] - IO.23
15 - EP1C6[74] - IO.22
16 - GND

Y6 : ANALOG
01 - ADCA - U34 - ADS7884[3] Input
02 - ADCB - U35 - ADS7884[3] Input
03 - DACA - U30 - DAC8830 - OP.U21[1] Output
04 - DACB - U27 - DAC8830 - OP.U21[7] Output
05 - AGND

III. Board Connectors

Z1 : micro USB
01 - VBUS
02 - DM - STM32F412[44]-PA11-USBDM
03 - DP - STM32F412[45]-PA12-USBDP
04 - ID - GND
05 - GND

Z2 : USB Type-C
01 - GND
02 - VBUS
03 - NC
04 - NC
05 - DP - GD32F103[24]-PA12-USBDP
06 - DM - GD32F103[23]-PA11-USBDM
07 - DP - GD32F103[24]-PA12-USBDP
08 - DM - GD32F103[23]-PA11-USBDM
09 - NC
10 - NC
11 - VBUS
12 - GND

Z3 : AUDIO JACK
01 - MIC
02 - HPCOM
03 - HPR
04 - HPL

Z4 : USB-A
01 - VDD_5.0V
02 - V3S[110] - DM
03 - V3S[111] - DP
04 - GND

Z5 : micro SD
01 - SSP_DAT2 - V3S[101]
02 - SSP_DAT3 - V3S[102]
03 - SSP_CMD - V3S[103]
04 - VDD_3.3V
05 - SSP_SCK - V3S[105]
06 - GND
07 - SSP_DAT0 - V3S[106]
08 - SSP_DAT1 - V3S[107]
09 - NC